

Social rank-order stability of mice revealed by a novel food competition paradigm in combination with available space competition paradigms

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eLife Assessment

This study proposes a **useful** assay to identify relative social ranks in mice incorporating the competitive drive for two basic resources - food and living space. Using this new protocol, the authors provide **solid** evidence of stable ranking among male and female pairs, while reporting more fluctuant hierarchies among triads of males. The evidence is, however, limited by the lack of ethologically based validation, assessment of the influence of competitor recognition, and proof of concept of application to neuroscience. This manuscript may be of interest to those interested in social behavior and related neuroscience.

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Abstract

Psychological, behavioral and biological studies on social organization and competition with animal models are boosting. The mouse has been recognized as a valuable and economic model animal for biomedical research in social behaviors, however, currently available food competition paradigms for mice remain limited. Discrepant paradigms involving different competitive factors, such as physical strength vs psychological features, muscular confrontation vs threat perception, and boldness vs timidity, may produce task-specific win-or-lose outcomes and lead to inconsistent ranking results. Here, we developed a food competition apparatus for mouse, in which contenders were a pair of mice eager to take over the same food pellet placed under a movable block in the middle of a narrow chamber where they were separated to the either right or left side. This Food Pellet Competition Test (FPCT)

was designed to (1) provide researchers with a choice of new food competition paradigm and (2) expose psychological factors influencing the establishment and/or expression social status in mice by avoiding direct physical competition between contenders. Meanwhile, we wanted to evaluate the consistency of social ranking results between FPCT and typically available space competition paradigms—tube test and warm spot test (WST). We hypothesized inconsistency of rankings of mice tested by FPCT, tube test and WST as they possess different targets for mice to compete and different factors determining competitiveness. Interestingly, application of FPCT in combination with tube test and WST discovered unexpected consistency of mouse social competitiveness and rankings in a grouped male or female mice that were housed in an either 2-or 3-member cage, most likely indicating that the status sense of animals is part of a comprehensive identify of self-recognition of individuals in an established social colony. Furthermore, the FPCT may facilitate researches on social organization and competition, given its reliability, validity, and ease of use.

Introduction

For gregarious animals and human beings, both competition and cooperation between individuals or populations are fundamental social behaviors important for the survival and evolution of the collective strains and species. Social competitions occur naturally to determine the ownership or priority of living territory, food, water, mates, other resources and non-resource requirements ^{1,2}. Through these competitions, social hierarchies from the dominant to the subordinate are established within a living group, where individuals at higher social status are granted with corresponding priorities of resources and non-resource requirements like living space, food availability, reproduction and safeguard ^{1,2}. The establishment of dominance hierarchies reduces the intensity and frequency of mutual aggression within groups and strains, so to maintain their inner stability and fulfil outer assignments ³. Impaired awareness of social competition has been documented in individuals with autism spectrum disorder (ASD) ^{4,5}, and reduced social interaction has been characterized in corresponding animal models ⁶. Similarly, maladaptive responses to social status loss has been associated with patient depressive disorders ^{7,8} and animal models of depression ^{9,10}.

Although humans and primates exhibit complex social interactions that are relatively easy to observe, their use for biological research is practically limited. In contrast, rodents —particularly mice and rats—have been emerging to serve as valuable and cost-effective models for studying the biological basis of social cognition, social interaction, and social organization ^{9,11,12}. Among the social competing objects, resources of living space, water and food are fundamentally essential for the survival of animate beings. Therefore, living space, water and food competitions are most frequently used to investigate social competitive behaviors and hierarchical ranking of rats or mice ^{2,13,14}. The tube test is a simple and reliable method for assessing social hierarchy by simulating the competition for living space among animals. In the tube test, the dominant mice largely consistently squeeze out the weaker ones from the narrow tube, indicative of an overwhelming space demand of mice at higher rank ¹⁵. Warm spot test (WST) is another space competition paradigm where a pair of mice compete for pitiful small warm corner in a freezing cage ^{12,16}. Some food competition tests have also been applied to animals such as rats ^{17–19}, chickens ²⁰, pigs ²¹, and mice ^{22–25}. Larger body size is an advantage of rats, chickens and pigs that makes detecting behavioral patterns easier by the experimenters or monitoring video, but the food competition procedures for those animals are difficult to be applied to mice due to its much smaller size. Both mice and rats are rodents and mostly used experimental animals, but rats are more socially tolerant and less hierarchical than mice ^{26,27}.

Relative to space competition, food competition tests for mice have been designated and applied less commonly in animal studies despite its long history ^{28–30}. Several issues could be thought to be the underlying limitations for the application of food competition paradigms. First, there are

methodological issues in some of these approaches, such as long video recording duration and difficulty in analyzing animal's behaviors during competitive physical interaction in videos, hindering their application by laboratories that cannot afford sophisticated equipment [23](#), [24](#), [31](#)–[35](#) and analysis [36](#). Second, aggressive behaviors often occur during competition in these approaches as the mice compete in a shared arena. Third, the prolonged food deprivation is usually required to increase the effortful food competition. For the second and third issues, on one hand, the aggressiveness/being aggressed and intense food deprivation cause stress responses and changes of physiological state of animals [10](#), [24](#), [28](#), [29](#). On the other hand, competition in a same arena conflates multiple determinants contributing to the outcomes of competition involving physical aspects—such as muscular strength, vigorousness of fighting, bite wounding—and psychological aspects—such as boldness, focused motivation, active self-awareness of status. This feature makes interpretation of the outcomes of competition difficult to discriminate the dominant behaviors and aggressive behaviors. So far, the food competition tasks had not been conceived to separate the physical aspects from psychological aspects interpreting the mice's winning/losing.

Therefore, to have a new choice of food competition paradigm for mice, and to facilitate the exposure of psychological aspects contributing to the winning/losing outcomes in competitions, we generated a convenient, easily operative and peaceful food competition paradigms for a pair of mice competing for a food pellet after mild calorie restriction. The FPCT was purposed for mice food completion without physical contact during competition, but tube test and WST were of space competition during which the mice need direct physical contact. Thus, we expected inconsistent evaluation results of competitiveness and rankings if we compared FPCT with typically available competition paradigms—tube test and WST.

Results

FPCT did not detect significant difference in the winning/losing results between unfamiliar non-cagemate male mice

The experimental procedure of FPCT consists of habituation, training and test (**Figure 1A**). After handling the mice in the homecage and experimental room, the mice were trained to enter the arena alternatively from the left and right gates to be familiar with the chamber arena (**Figures 1B**, **S1** and **S2**), followed by next step to find a small food pellet in the trough in the middle of the chamber floor (**Figures 1C** and **S2**). At the end of this step of training, the mice would go directly for the food pellet when they entered the arena (Video **S1**). Then, a movable block was hung up at the roof of the chamber (**Figures 1D**, **S1** and **S2**). The bottom of the high rectangular block was positioned nearly right above the trough so that the mice could not possess the food pellet unless they pushed the block away. The block is transparent and dug with holes at the lower part, enabling the mice see and smell the food pellet (**Figure 1D**). The mice were trained to learn to move away the block and reach the food pellet twice daily (**Figure 1D**). Usually, it took 4–5 days for the mice to learn to walk straightforwardly to the obstacle once they entered the chamber and move the block easily to expose the food pellet (**Figure 1E** and Video **S2**). Orientation of the entries did not generate bias of performance of mice in the apparatus (**Figure 1E**).

In the test section, two mice were allowed to enter the separated rooms of the chamber simultaneously from one of the two entries, competing for the food pellet either placed in the trough below the block (test 1, **Figure 1F**)—where the pallet was accessible if the block was pushed away—or on the inner bottom of the block (test 2, **Figure 1G**)—where the pallet was inaccessible. The two mice were able to see and sniff each other through the transparency and holes of the block, as well as the gap between the block and chamber.

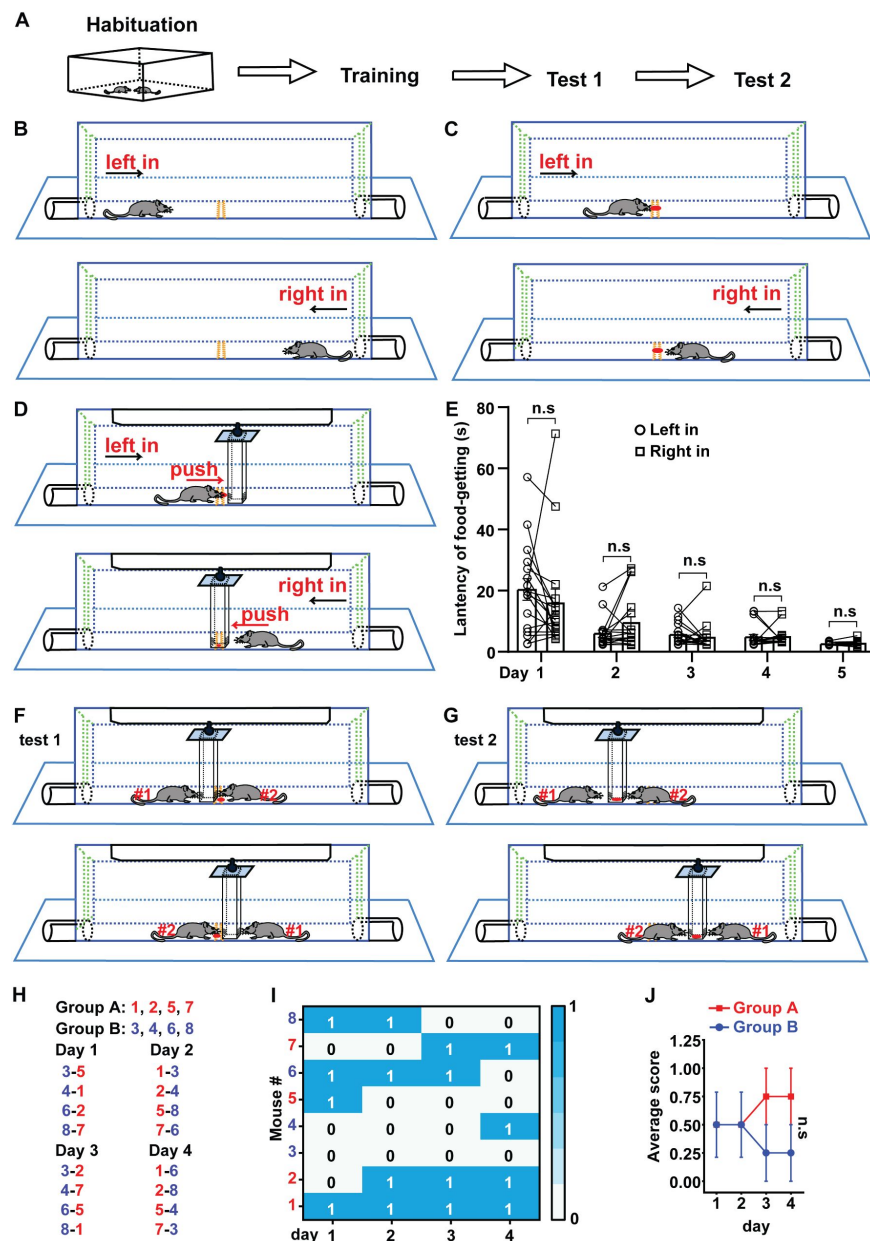


Figure 1.

Schematic drawing of FPCT procedure and measuring the non-cagemate male mice based on the winning/losing outcomes of FPCT.

(A) The overall procedure of FPCT experiment. (B) Habituation to the arena where the mice entered alternately from left and right sides. (C) Training to find food pellet without the existence of the movable block. (D) Training to get the food pellet placed under the movable block. (E) Statistical data of latency of food-getting (the time from entering the arena to eat the food) showing the progress of training in (D). paired Student's *t* test, *n* = 16, n.s. stands for non-significant difference. (F) Direct win-lose test via analyzing food occupation (test 1). (G) Indirect win-lose test via analyzing mouse's attempt to gain the food (test 2). (H) Eight single-housed mice were numbered and randomly divided into group A and B after being trained. One of the mice in group A matched only once with one of the mice in group B in the 4 days of test 1. The number listed on the left-right indicated the corresponding mice would enter the arena from left-right entry. Score 1 and 0 indicated winning and losing the food competition, respectively. (I) Heatmap showing the outcomes food competition of the non-cagemate male mice ranked with FPCT. (J) Average score of the two groups acquired in the competition. Unpaired Student's *t* test = 4, n.s. stands for non-significant difference.

In the first experiment utilizing the FPCT, we tested age- and weight-controlled unfamiliar non-cagemate male mice to examine whether there was obvious non-social priority-associated competitiveness expressed in the FPCT. Eight single-housed mice were trained and then randomly divided into groups A and B for competition test 1. Within the 4 consecutive days of competition test there were totally 16 trials, as one mouse in a group matched one of mice from the other group once daily and the matched two players participated in the contest only once in the 4-day schedule (**Figure 1H**). In each trial of the test 1, the mouse which obtained the food pellet was credited a score of 1 (time of winning), while the mouse which failed to obtain the food pellet was credited a score of 0 (**Figure 1I**). Statistical data found similar average scores acquired by the two groups of mice (**Figure 1J**), illustrating that no non-social priority-associated competition factor was obviously involved in the FPCT with the sex-, age- and weight-controlled unfamiliar mice.

FPCT identified different social ranks between 2-cagemate male mice, verified by tube test

Social hierarchy or status of many animals is largely arranged by the outcomes of winning/losing social competitions, manifesting as dominant or subordinate behaviors in a well-established social colony. To see the efficacy of FPCT in detecting the social status of mice in a well-established social colony, we conducted 4-day of FPCT test 1 containing 1 trial per day for each of the 16 pairs of cagemate male mice after training (**Figure 2A**). In the neighboring 2 days, each mouse was allowed to enter the chamber from different entries. In each trial, the mice were scored 1 for winning and 0 for losing the food pellet. In the final trial, the mouse obtaining the food pellet was ranked #1 and considered as the final winner, while the mouse failing to get the food pellet in the match was rank #2 and claimed as the final loser. Retrograde analysis of the time spent to get the food after entering the arena gate in the last day of training showed that the winners and losers displayed same level of training to get the food and equivalent motivational craving for the food (**Figure 2A**). However, when there was competition in the test the winners tended to win the pellet overthought all the 4 trials (**Figure 2A**). Notably, the presence of an opponent during the test did not significantly change the latency to get the food, comparing with last day of training when there was no opponent (**Figure 2A**). In all the 16 pairs of mice, only 1 pair exhibited alternating winning/losing outcomes with an inter-trial consistency at 50% (**Figure 2B**), while all the other 15 pairs kept 100% consistency of winning/losing readouts (**Figure 2C**). Averagely, the competition results were stable throughout the 16 trials (**Figure 2D**) and a rate of consistency reached 96.88% (**Figure 2E**). In the test 2 (indirect win-lose test via analyzing mouse's attempt to gain the food), we observed that the winners determined by the final trial of test 1 tried harder to get the food pellet by spending more time to push the block within the single 2-minute test, a limited time to avoid distinction of attempt (**Figure 2F**). Thus, these data demonstrate that FPCT are effective to rank the established social status of mice, and that consecutive 4 trials (one trial daily) might be least required for the FPCT to achieve readout consistency and reliability.

Many factors may contribute to the competitive outcomes in mice, such as age, sex, training level, physical strength, and intensity of psychological factors. We were interesting to understand which factor determines the outcomes of mice's winning/losing in the FPCT match. First, effects of age and sex on winning/losing outcomes were excluded as the mice were age-matched and solely males. Second, the weight of mice was also excluded to determine the competency possibly by providing physical strength of pushing, as the weight difference between paired contender mice before training was controlled in less than 10%, and no difference between the winners and losers in weight measured again after finishing the food competition tests (**Figure 2G**). Third, biased training level should produce distinct likelihood of winning and losing, but retrospective comparison of the duration they took to obtain the food pellets in the last day of training when there was no competition reported no difference in the latency to get the food pellet between the winners and losers (**Figure 2A**), indicating that the mice were equally trained and motivated. Forth, the major physical contribution to the competition based on violence aggressiveness and

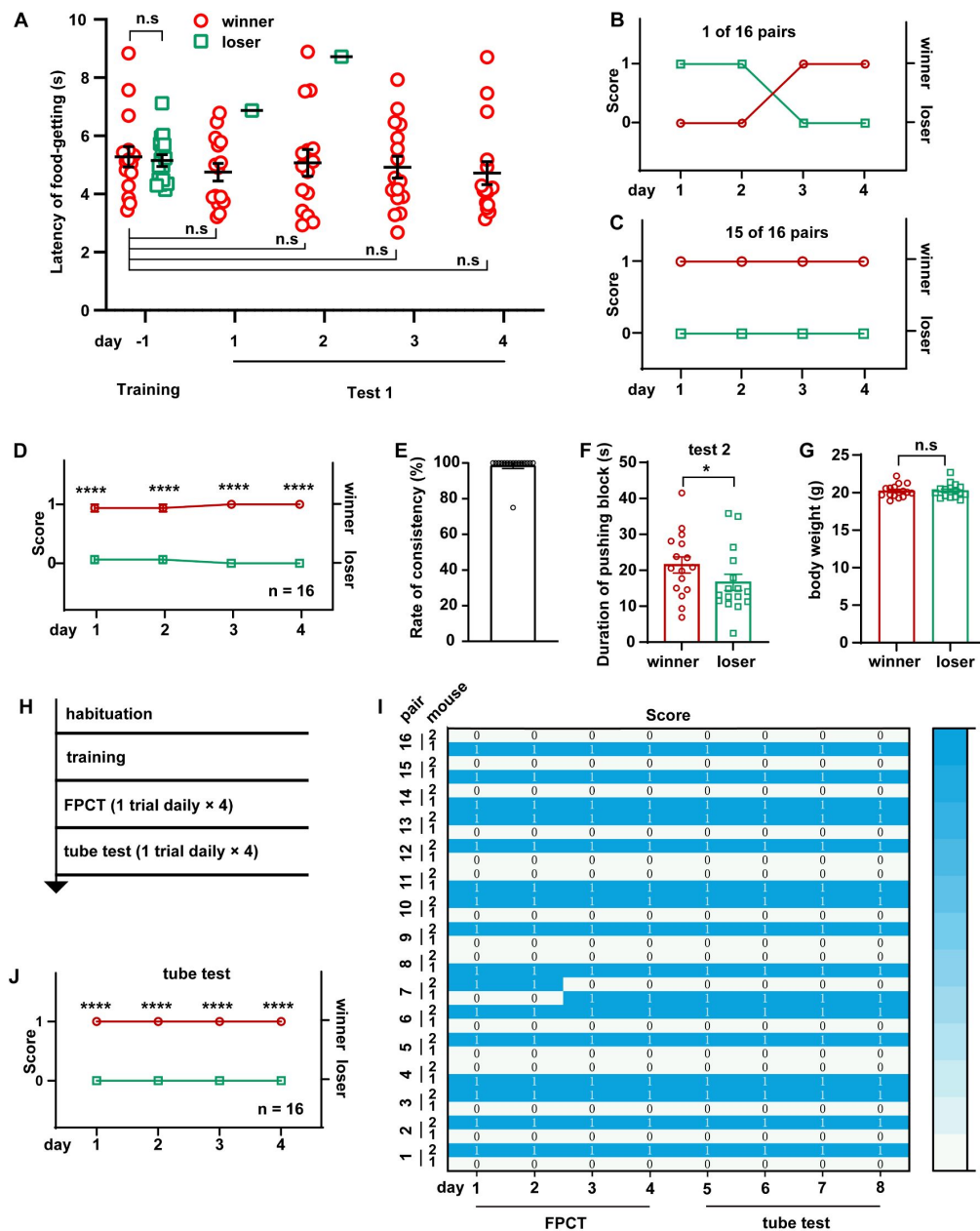


Figure 2.

Ranking the 2-cagemate male mice with FPCT and tube test.

(A) Statistical data showing the latency of food-getting from the last day of training until the end of the test 1. Two-way ANOVA multiple comparisons, $n = 16$ pairs, n.s. stands for non-significant difference. (B) 1 of the 16 pairs of cagemate male mice exhibited alternating winning/losing outcome in FPCT test 1. (C) 15 of the 16 pairs of male mice exhibited fully congruent winning/losing outcome in FPCT test 1. (D) Statistics of ranking of male mice over through 4 trials (1 trial daily) of FPCT test 1. Two-way ANOVA, $n = 16$, **** $P < 0.0001$. (E) Average rate of consistency within trials. For each pair, rate of consistency was calculated as the percentage of the number of trials (in all 4 trials) resulting in the outcome same as the 4th trial ($n = 16$). (F) Statistics of duration spent on pushing block in FPCT test 2. The winner or loser identity was determined by the last trial of test 1. Paired Student's t test, * $P < 0.05$, $n = 16$. (G) Body weight of male mice measured after FPCT. Unpaired Student's t test, n.s. stands for non-significant difference, $n = 16$. (H) Timeline of experiments showing tube test was conducted after FPCT. (I) Heatmap showing the outcomes of social competition of paired male mice ranked with FPCT and tube test. (J) Statistics of ranking of male mice over through 4 trials (1 trial daily) of tube test. Two-way ANOVA, $n = 16$, **** $P < 0.0001$.

fighting strength were minimized in this competitive scenario due to separation of the mice by the obstacle. Thus, we postulated that the winning/losing outcomes in this paradigm mainly rely on mice's state of psychological factors, for example, intensity of motivation, self-awareness of social status, memory of winning/losing experience in the home cage, fear of revenge when returning the living cage. Most possibly, self-awareness of social status plays a determinant role in the outcomes.

To see whether social ranking revealed by the FPCT is consistent with the tube test that represents the simple and robust behavioral assay for space competition ¹⁵, we continued the experiments to rank the mice using tube test following FPCT (**Figures 2H** and **2I**). We found that consistency of outcomes between the FPCT and tube test were 100 percent, as the winners and losers in the last trial of the FPCT continued consecutively to be the winners and losers, respectively, during the 4 consecutive trials of tube tests (**Figures 2I** and **2J**). Therefore, the rank-order examined by with the FPCT is fully verified by the tube test.

Social ranking of 2-cagemate female mice using FPCT, tube test and warm spot test

It is widely accepted that males, including human beings and animals, are evolutionarily more eager to be dominant, more aggressive and more hierarchical, but it is arguable regarding whether females have less competition and looser social organization ^{37–39}. On the base of investigation of aggressive behaviors, it has been reported that the social colony of female mice is also hierarchical ^{30,40}. Taking use of this newly designed mouse competition task in which dependence of competency on physical aggressiveness is minimized, we examined whether intra-female social status between two cagemate female mice was observable in the FPCT context. Notably, the FPCT test 1 showed that 3 in 10 pairs of cagemate female mice exhibited alternating winning/losing outcomes, while the majority of cagemates (7 in 10 pairs) showed fully congruent grades (**Figures 3A–3E**). Averagely, the 10 female pairs displayed stable rankings over through all trials (**Figure 3F**). The overall rate of inter-trial consistency was 90% (**Figure 3G**), which was not significantly different from male mice ($P > 0.05$ comparing data in the **Figures 2E** and **3G** by Mann-Whitney U test). In the test 2, we observed that winner mice in the test 1 paid more efforts trying to get the food pellet (**Figure 3H**). The body weights examined after the tests (**Figures 3I**) were similar between the winners and losers, so the two parameters were not likely to contribute to the competency. Interestingly, retrospect analysis of training data displayed similar training level of food-getting and craving state for food (**Figures 3A**). However, the presence of an opponent during the test 1 could significantly disturb the female mice and increase the its latency to get the food, comparing with last day of training when there was no opponent (**Figure 3A**). These FPCT results suggest that the social relationship of female mice is also stratified.

In addition to the tube test, WST, the regional urine marker and the courtship ultrasound vocalization test are established behavioral methods for determining social rank in mice ^{12,14,16,17,32,41–46}. Tested animals in different tasks may pursue specific goals, and utilize their specific expertise, strength and skills. Thus, we assumed that different competition paradigms may produce inconsistent social ranking readouts of the same social colony, resulting from various types and levels of competition. To check this assumption, we continued to assess these female mice using tube test and WST following FPCT (**Figure 3J**). As expected, some of the trials exhibited inconsistent rankings the three tests. However, most of the trials unexpectedly showed that mice maintained the winner or loser identity acquired in the FPCT in subsequent tube test and WST (**Figures 3K–3M**). Overall, no difference was observed in the rate of inter-trial consistency comparing FPCT with tube test, FPCT with WST, and tube test with WST (**Figure 3N**). These data illustrate that mouse social competency and status of established female mice colony are overall stable so that a well competitive subject exhibits dominance not just in a specific, but in a variety of contexts, despite that the different contexts contain distinct competitive factors like food and space, or competitive aspects like physical strength and psychological factors.

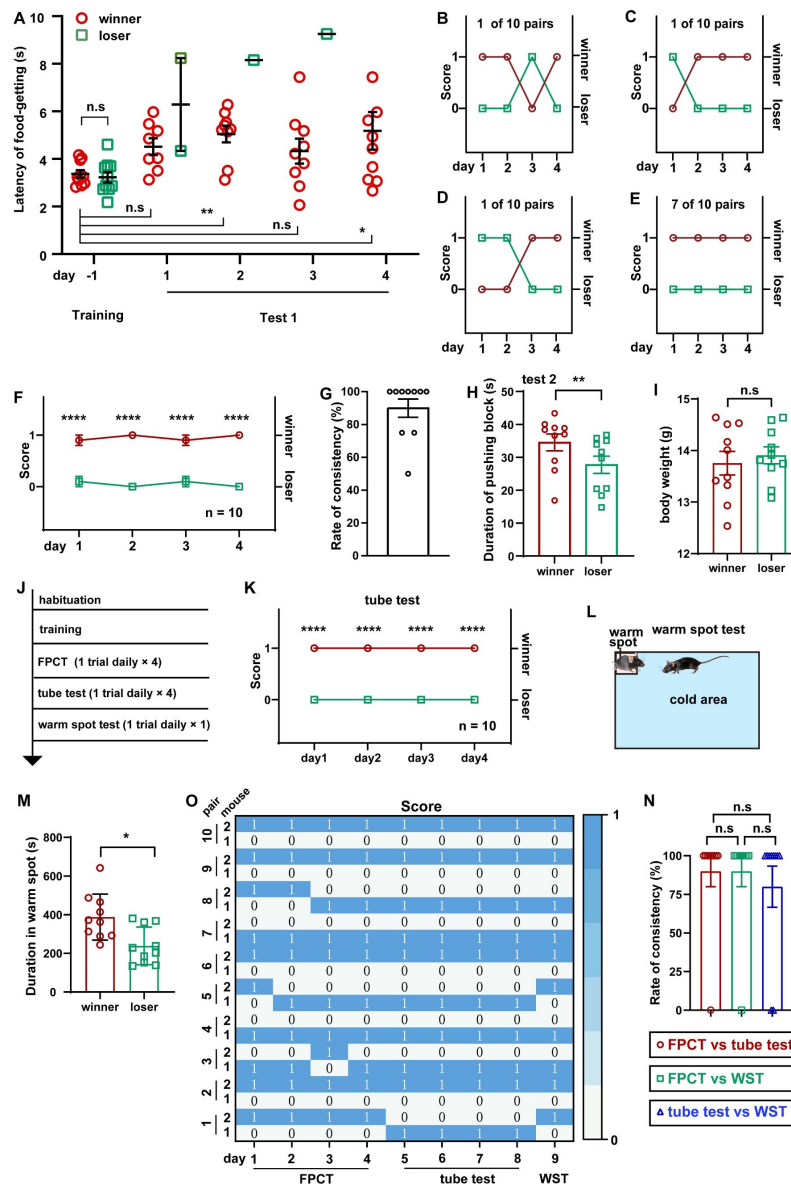


Figure 3.

Ranking the 2-cagemate female mice using FPCT, tube test and warm spot test (WST).

(A) Statistical data showing the latency of food-getting from the last day of training until the end of the test 1. Two-way ANOVA multiple comparisons, $n = 10$ pairs; $*P < 0.05$, $**P < 0.01$, n.s. stands for non-significant difference. (B-D) 3 of the 10 pairs of cagemate female mice exhibited alternating winning/losing outcome in FPCT test 1. (E) 7 of the 10 pairs of female mice exhibited fully congruent winning/losing outcome in FPCT test 1. (F) Statistics of ranking of female mice over through 4 trials (1 trial daily) of FPCT test 1. Two-way ANOVA, $n = 10$, $****P < 0.0001$. (G) Average rate of consistency within trials. For each pair, rate of consistency was calculated as the percentage of the number of trials (in all 4 trials) resulting the outcome same as 4th trial ($n = 10$). (H) Statistics of duration spent on pushing block in FPCT test 2. The winner or loser identity was determined by the last trial of test 1. Paired Student's t test, $*P < 0.05$, $n = 10$. (I) Body weight of female mice measured after FPCT. Unpaired Student's t test, n.s. stands for non-significant difference, $n = 10$. (J) Timeline of experiments showing tube test and warm spot test were conducted after FPCT. (K) Statistics of ranking of female mice over through 4 trials (1 trial daily) of tube test. Two-way ANOVA, $n = 10$, $****P < 0.0001$. (L) Schematic of the warm spot test. (M) Cumulative duration of mice occupying the warm spot. The winner or loser identity was determined by the last trial of FPCT. Paired Student's t test, $n = 10$, $*P < 0.05$. (O) Heatmap showing the outcomes of social competition of paired female mice ranked with FPCT, tube test and WST. (P) Rate of consistency between FPCT and tube test (day 4 vs day 5), FPCT and WST (day 4 vs day 9), as well as tube test and WST (day 8 vs day 9). One-way ANOVA test, $n = 10$, n.s. stands for non-significant difference.

Social ranking of triads of male mice using FPCT, tube test and WST

Social organization of the bigger crowd could be much more complicated than the simplest 2-member group. It is controversial whether the dominate and subordinate roles in a larger group of mice tend to be mobile or rigorous in a variety of contexts^{35,47,48}. To probe this issue, we raised 3 male mice in a cage and then following adequate habituation and training we ranked their hierarchies using FPCT, tube test and warm spot test sequentially (**Figure 4A**). In the FPCT and tube test, the 3 cagemate mice contested in a round-robin style as the both were designated for two contenders in a time. The ranking result showed that 6 in 9 groups of mice displayed some extent of flipped ranking (**Figures 4B–4G**), and only 3 in 9 groups displayed continuously unaltered ranking (**Figure 4H**). Averagely, in the totally 27 trials consisting of 12 trials of FPCT, 12 trials of tube test and 3 trials of WST, an obvious stable linear intragroup hierarchy was observed across all the trials and tasks (**Figures 4I** and **4J**). Comparison of inter-task consistency revealed that the ranks assessed by FPCT, tube test and WST did not differ from each other (FPCT vs tube test, $85.19 \pm 17.95\%$; FPCT vs WST, $81.59 \pm 17.95\%$; tube test vs WST, $81.48 \pm 17.95\%$; $P > 0.05$, one-way ANOVA test. **Figure 4K**). These results, together with the results of female mice tested by FPCT, tube test and WST, illustrate that the dominate and subordinate roles in an established group of mice are rigorously linearly stratified so that the status can be manifested in various contexts.

Discussion

Social hierarchy is an innate feature of gregarious animals, usually established through social competition. Among the diverse contents for social competition, resources of living space and food are fundamentally critical priority for animals^{2,13}. Mice have been designated to participate in food competition tests for a long time, since at least the 1950s^{28,29}. In these applied food competition tests, social ranks were scored by calculating the total amount of food consumed by each mouse competing in the same chamber or cage, and/or analyzing their physical aggressiveness^{22,23,30,48}. Some common limitations for the wide distribution of these food competition tests are significant, including the occurrence of aggressive behaviors, requirement of prolonged food deprivation, calculation formulas, long video recording duration, and difficulty in discriminating individual animal's behaviors during interacting with group members in videos^{23,24,31–35}. Food shortage serves as a strong driver to induce aggressiveness and competitive behaviors⁴⁹. However, in rodents, being physically aggressed and undergoing intense food deprivation can influence social competitive behaviors via disturbing the internal state^{24,31,50} and triggering robust stress responses^{49–53}.

Here, we developed and validated a food competition assay—FPCT, a simple and effective tool designed to provide a means of assessing social status of mice, favoring the growing scientific interest in understanding the neurobiological insights of social hierarchy. The potential influence of food restriction on competitive behavior was minimized in our task where the mice underwent only a 24-hour food deprivation period at the beginning of training, followed by restrict food supply to meet basic energy requirement. The aggressive situations, at least direct physical fighting, was prevented during this ranking test due to the separation of the mice in either side of the arena. The FPCT paradigm is conveniently and generally applicable for its simple procedure. First, the mice were housed together to habituate each other and establish hierarchical social structure. Then, the mice were trained individually to accommodate the chamber arena, know the existence and position of the food pellet, and learn to push away the block above the food pellet to get the food. Importantly to note, stable social hierarchy relies on adequate habituation and reliable ranking test is based on well training. Finally, the mice underwent the ranking test in the chamber, where they were allowed to enter from either the right or the left side of the chamber separated by the movable block to compete for the same food pellet under the block.

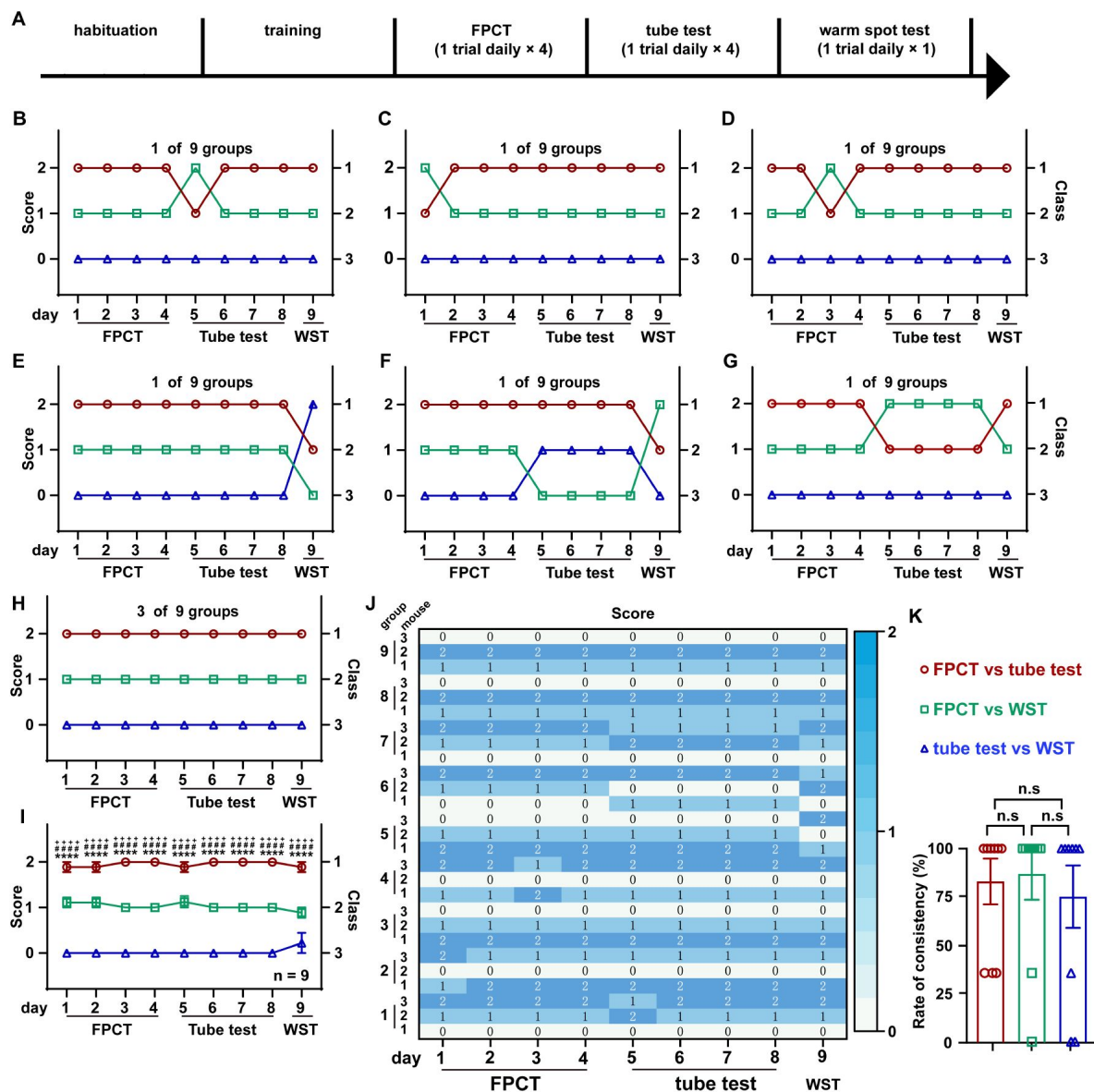


Figure 4.

Ranking the triads of male mice using FPCT, tube test and WST.

(A) Timeline of experiments to rank 3-cagemate male mice using FPCT, tube test and WST sequentially. In the FPCT and tube test, the mice contested in a round-robin style within the each 3-cagemate group. (B-G) 7 of the 10 groups of male mice exhibited alternating winning/losing outcome during the whole competition tasks. (H) 3 of the 10 groups of male mice exhibited fully congruent winning/losing outcome during the whole competition tasks. (I) Statistics of ranking of mice over through the whole competition tasks. Two-way ANOVA test, $n = 10$, **** $P < 0.0001$ comparing FPCT and tube test, **** $P < 0.0001$ comparing FPCT and tube test, **** $P < 0.0001$ comparing tube test and WST. (J) Rate of consistency between FPCT and tube test (day 4 vs day 5), FPCT and WST (day 4 vs day 9), as well as tube test and WST (day 8 vs day 9). One-way ANOVA test, $n = 10$, n.s. stands for non-significant difference. (K) Heatmap showing the ranking outcomes of grouped male mice during the whole competition tasks.

Concerning the interpretation of the ranking results, both static vs dynamic and linear vs despotic hierarchy organizations of grouped mice that have been documented ^{35,47,48}. The diversity of ranking results might be attributed to various factors. For example, different paradigms containing different tasks that may enable different mice to express its proficient competitiveness such as physical advantage, psychological strength, and learning ability. Bodily conflict between contender mice in a shared space could make muscular strength be competitive. It is not sure if body size/weight affect social hierarchy as both correlation ³⁵ and non-correlation ^{10,48} between weight and hierarchy have been reported. Strategic task possibly promotes a smarter mouse to be a winner in the test even if it may in a subordinate role at homecage. Besides, group size of experimental animals ^{54,55}, duration of habituation of group members ⁵⁵, training level and test protocols (e.g., short-term vs long-term test) ⁴⁷ should be considered as common points give rise to instability of the formation or expression of social hierarchy. In this novel food competition paradigm, we tested pairs and triads of same-sex mice with similar age and body weight, housed in the same cage for at least 3 days. We did not find significant difference in food priority within pairs of non-cagemates, whereas we detected overall stable winning/losing outcomes of cagemate pairs and triads. Considering that direct bodily competition between mice was avoided in the FPCT by separating the matching pair in either side of the chamber, the stable ranking results of the matches should be mainly determined by the psychological aspects of the contenders. Most possibly, it is a reflection of mouse's self-awareness of social status.

Combining FPCT with typically available paradigms—tube test and WST, we addressed 3 questions: whether female mice are stably socially structured, whether FPCT is effective to discover hierarchy rankings across multiple cagemates, and whether different social tasks result in discrepant hierarchy rankings. First, we found that the majority of female pairs displayed overall stable hierarchy ranks, consistent with previous findings ^{40,56}. We also detected 3 in 10 female cagemate pairs exhibiting fluctuant rankings during the 4-day FPCT test, but it was not significant from 1 in 16 male pairs. It has been revealed by agonistic behaviors in previous work that female social hierarchies are not as steep and despotic as male hierarchies ³⁹. The discrepancy between this work and ours might be attributed to difference in mouse strains, group sizes, housing models and detecting methods. Second, by means of round-robin tests of multiple cagemates, both FPCT and tube test detected averagely linear intragroup hierarchies, consistent with each other. Third, considering that diverse competitive factors lying in different tasks, the ranking readouts of male pairs and triads examined by FPCT, tube test and WST were highly consistent with each other, revealing social rank-order stability of mice across the three types of examining tasks. Notably, the fluctuance of rankings occurs occasionally, suggesting that the social status of mouse could be dynamic and transitive in some occasions ⁴⁷. It would be interesting to investigate whether the social hierarchies in larger sizes of mice group, especially female mice, are more dynamic and transitive. Alternately, the occasional intra-trial and inter-task ranking fluctuations might be a result of incompetency of paradigms to reflect the real intra-colony structure of mice.

In conclusion, our data suggest that hierarchical sense of animals might be part of a comprehensive identify of self-recognition of individuals within an established society. Hopefully, the FPCT will facilitates future studies to reveal more detailed properties of social organization, social competition and their underlying neurobiological mechanisms.

Methods and materials

Animals

All animal experiments were conducted according to the Regulations for the Administration of Affairs Concerning Experimental Animals (China) and were approved by the Southern Medical University Animal Ethics Committee. The mice used in this study were 6-8 weeks old C57/BL6J

purchased from SPF (Beijing) Biotechnology Company (Beijing, China) and were raised in the environment with the relative humidity 50-75%, temperature 22-24°C and 12-12-h light-dark cycle. The mice were allowed to freely access to food and water unless undergoing food restriction requirement during experiments.

Food pellet competition test (FPCT) setup

The experimental setup consists of a food competition arena (Figure S1) and a camera. The arena includes a base, a chamber, two entries, two doors, a movable block, a pulley system, a food trough and the roof (Figure S2) made up of a batch of wheels and a wheel track for the pulley system, as well as 15 pieces of acrylic plates for other parts listed with dimensions in Figure S2F. The movable block hung up under the track section separates the arena into the left and right compartments. It is necessary to apply some paraffin oil to the track to reduce the friction between the pulley and the track (Figure S3). The food trough is positioned in the middle of the chamber floor and nearly under the movable block unless the block is pushed away by mice. Food pellets are tiny delicious milk crackers. Mouse behaviors are monitored by a side view camera with PlotPlayer software.

FPCT procedure

The experimental procedure consists of habituation, training and test (Figure 1A [↗](#)). The details of the procedure are described below.

Habituation

Contender pair of C57BL/6JNifdc mice with similar age and weight (the weight difference was less than 10%) were housed in the same cage for at least 3 days under a 12-12-h light-dark cycle. Before the formal training of mice, each mouse was handled by experimenters for 3-5 minutes per day for approximately 3 days. Overall, the habituation period lasts about 6-8 days.

Training

Step 1, food restriction. Food restriction includes food deprivation and food limitation. First, the mice were deprived of food for 24 hours while water consumption remained normally to enhance the appeal of the food pellet to the mice. Then, after 24 hours of food deprivation, each cage of mice was given 10 g of food every morning to meet their daily food requirements until the end of the test.

Step 2, contextual familiarization (Figure 1B [↗](#)). At each time after mice homecage was translocated from animal facility to behavioral room, the cage lid was removed to allow the mice to freely explore the cage for about 1-2 minutes to become accustomed to the open roof. Before training to find the food pellet (step 3), each mouse was allowed to enter the arena chamber from the left side where the mouse was kept for 3 minutes before been gently driven out from the right side of the chamber to be back to homecage to have a 2-minute rest. Then, the mouse would enter the arena chamber from the right side and 3 minutes later was gently driven out from the left side. Arena familiarization was conducted 1 round per day for 2-3 days.

Step 3, training to find the food pellet (Figure 1C [↗](#)). The experimenter placed a small food pellet in the trough in the middle of the floor of chamber. Let one mouse enter the chamber from the left side and stay there until it found and ate the pellet. After that, the mouse was gently driven out of the chamber from the right side to have a 2-minute rest in its homecage. Then, the mouse would enter the chamber from the right side and stay there until it found and ate food. The training was repeated 1 round daily until each mouse directly went to the trough and ate the small pellet after entering the chamber (Video S1).

Step 4, learning to get the food pellet (**Figure 1D** [↗](#)). In this step, the small food pellet in the trough was nearly under the vertical block. As the block was transparent and had holes at lower portions facing to the entries, and it was movable thanks to the sliding wheels and the track at the top of the block attached to the roof, the mouse was able to learn to push away the block to get the pellet. The roof attached with track and block was lifted up after the mouse ate the pellet to allow the mouse go out of the chamber from the side opposite to the entry side. Each mouse was repeatedly trained to get the food pellet alternatively from left and right entry until they were able to harvest the pellet directly and easily without competition (Video S2).

Test

Test 1, direct win-lose test via analyzing food occupation (**Figure 1F** [↗](#)). Test 1 required several trials, in the first of which the paired contenders entered the chamber from the opposite sides simultaneously. The mice entered the chamber from left or right entry alternatively in two consecutive trials. As there was only one small pellet placed under the movable block in each trial, the mouse that obtained the pellet was deemed as winner, while the other one was the loser (Video S3). To calculate the latency of food harvesting of mice, a starting state was marked when its four legs just entered the chamber arena (Figure S4). The time spent from starting state to food-getting for a mouse was calculated as latency of food harvesting. One competition test comprised at least 4 consecutive trials.

Test 2, indirect win-lose test via analyzing mouse's attempt to gain the food (**Figure 1G** [↗](#)). In this test, a slightly larger pellet was placed in the inner bottom of the block, rather than in the trough under the block, so that the mice could see, smell but not access it. Once the mice entered the chamber, they would push the block and attempted to gain the food. The winner or loser was determined by times of pushing the block in 2 minutes (Video S4).

Tube test

The detailed steps of tube test were described previously by Fan et al. ¹⁵ [↗](#). It consists of the following three steps: adaptation, training and test. The adaptation phase had been completed at FPCT, referred to as the habituation step of FPCT. In the training phase, the tail of the mouse was gently lifted and placed at one end of the tube, and when the mouse entered the tube, the tail was released and the mouse was allowed to pass through the tube. A plastic rod was used, when the mouse retreated or stagnated for a long time, to gently touched the tail so that the mouse could continue to move until it passed through the tube. The test phase required four consecutive days, and the social rank of the mice was ranked for four consecutive days. During the experiment, the camera was placed directly in front of the tube, and the whole process of the mouse experiment could be recorded.

Warm spot test (WST)

The experimental apparatus utilized for the WST consisted of a rectangular behavior box with dimensions of 28 cm in length, 20 cm in width, and 40 cm in height. Prior to the commencement of the experiment, the bottom of the box was placed on ice to ensure that its temperature was approximately 0°C. A heating sheet measuring 2 cm by 2 cm was positioned in one of the inner corners of the box, providing sufficient space for only one mouse. The temperature of the heating sheet was maintained at 34°C, and this temperature was monitored using a temperature measuring gun. During the experiment, three mice were introduced into the box simultaneously, allowing them to move freely for a duration of 15 minutes. The entire experimental procedure was recorded via video, and the time each mouse spent on the heating sheet was subsequently quantified from the recorded footage.

Statistical analysis

The experimental data were statistically analyzed using *Prism* 9 or origin 9.0 software and presented as mean $\pm$ SEM. Two-way or One-way ANOVA, unpaired or paired Student's t test and Mann-Whitney U test were used to compare the difference between groups. Statistical significance level (*P* value) was set at 0.05.

Supplementary figures

A



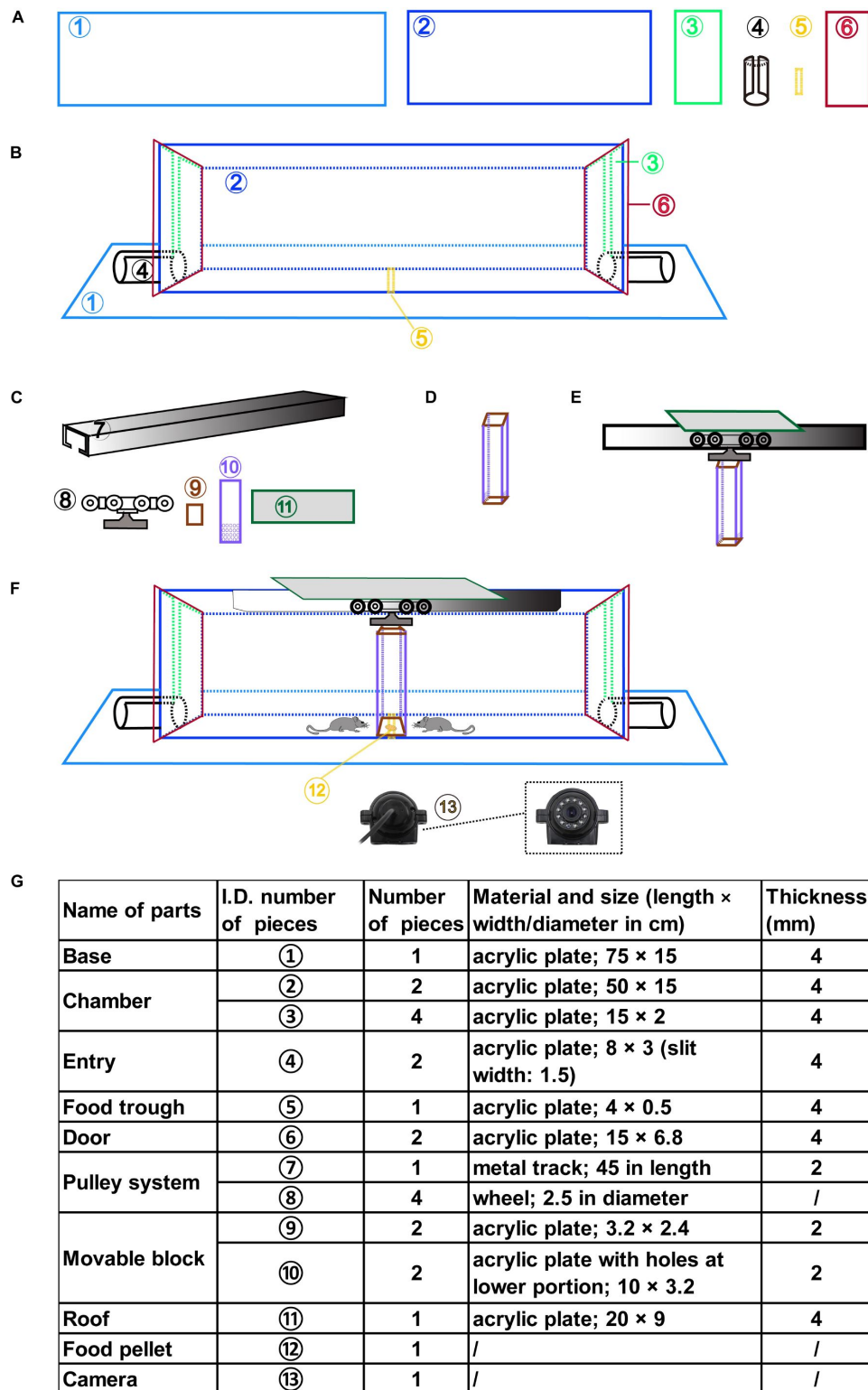
B



Supplementary figure 1 (Figure S1).

Photographs of food pellet competition test (FPCT) setup.

(A) Frontal view of FPCT setup. (B) Side view of FPCT setup.



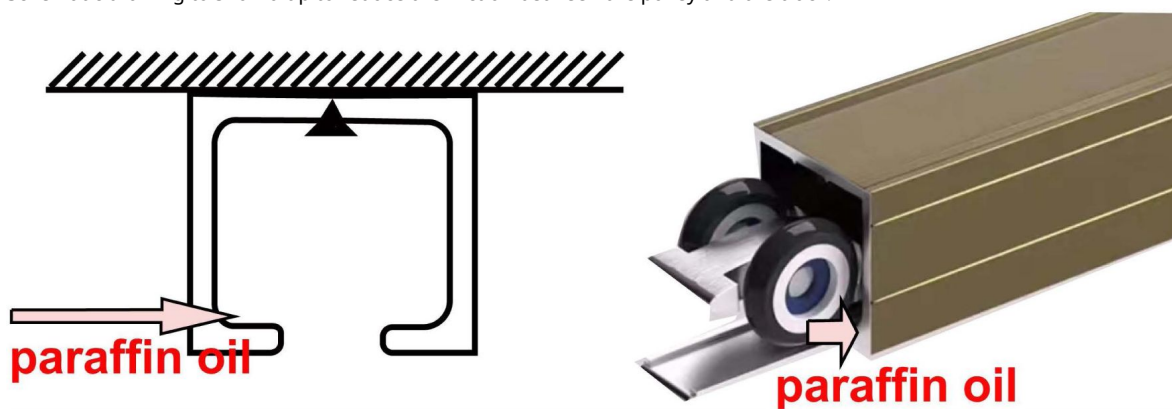
Supplementary figure 2 (Figure S2).

Schematic illustration of the assembling of the FPCT setup.

(A-B) Depiction of the work pieces (A) to make up the main body of the arena (B). (C-E) Depiction of the work pieces (C) to make up the movable block (D) attached to the pulley system and the roof (E). (F) Cartoon demonstrating the FPCT setup and working model with the existence of food pellet and camera. (G) Detailed list of the work pieces and parts of the FPCT setup.

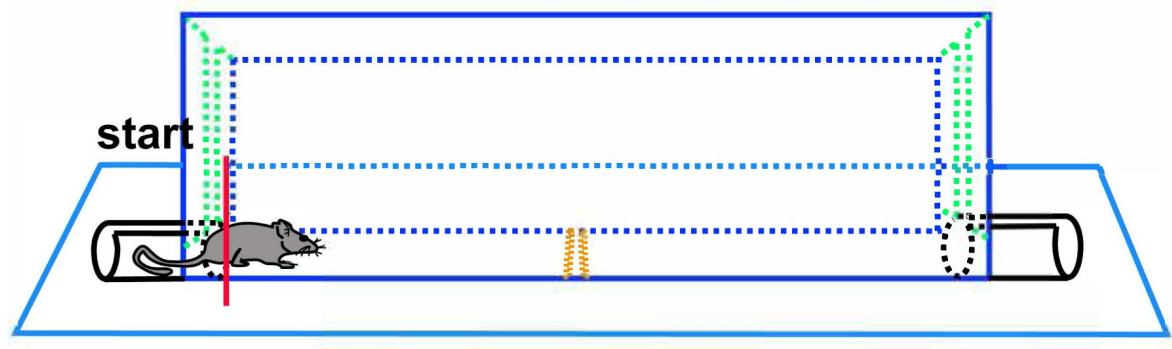
Supplementary figure 3 (Figure S3).

Schematic drawing to show a tip to reduce the friction between the pulley and the track.



Supplementary figure 4 (Figure S4).

Schematic drawing to show the marking of the starting state when the mouse's four legs just entered the chamber arena.



Additional information

Author contributions

M.L. and Y.C. acquired the experimental data. M.L. analyzed the data. M.L. and R.C. established methodology and wrote the manuscript. R.C. conceived the project and supervised the research. All authors contributed to the finalization and approved the content of the manuscript.

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Availability of data and materials

The data and supplementary information supporting the findings of this study are within the paper and other information helping to understand this study could be available upon request from the corresponding authors.

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Additional files

Video S1. *In the step 3 of training, the mouse was trained to know the position of the food pellet in the absence of the movable block.* [🔗](#)

Video S2. *In the step 4 of training, the mouse was learning to get the food pellet placed in trough of the chamber floor nearly under the movable block.* [🔗](#)

Video S3. *In FPCT test 1, the mice were competing for the food pellet.* [🔗](#)

Video S4. *In FPCT test 2, the mice were attempting to get the food pellet placed in the inner bottom of the block.* [🔗](#)

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Reviewer #1 (Public review):

Summary

The authors present a new protocol to assess social dominance in pairs and triads of C57BL/6j mice, based on a competition to access a hidden food pellet. Using this new protocol, the authors have been able to identify stable ranking among male and female pairs, while reporting more fluctuant hierarchies among triads of males. Ranking readout identified with this new apparatus was compared to the outcome obtained with the same animals competing in the tube and in the warm spot tests, which have been both commonly used during the last decade to identify social ranks in rodents under laboratory conditions.

Strengths

FPCT allows for an easy and fast identification of a winner and loser in a context of food competition. The apparatus and the protocol are relatively easy and quick to implement in the lab and free from any complex post processing/analysis, which qualifies it for wide distribution, particularly within laboratories that do not have the resources to implement more sophisticated protocols. Hierarchical readout identified through the FPCT correlates with social ranks identified with the tube and the warm spot tests, which have been widely adopted during the last decade and allow for study comparison.

Weaknesses

While the FPCT is validated by the tube and the warm spot test, this paper would have gained strength by providing a more ethologically based validation. Tube and warm spot tests have been shown to provide conflicting results and might not be a sufficient measurement for social ranking (see Varholik et al, Scientific Reports, 2019; Battivelli et al, Biological Psychiatry, 2024). Instead, a general consensus pushing toward more ethological approaches for neuroscience studies is emerging.

Other papers already successfully identified social ranks dyadic food competition, using relatively simple scoring protocol (see, for example, Merlot et al., 2006), within a more naturalistic set-up, allowing the 2 opponents to directly interact while competing for the food. A potential issue with the FPCT, is that the opponents being isolated from each other, the normal inhibition expected to appear in subordinates in presence of a dominant to access food, could be diminished, and usually avoiding subordinates could be more motivated to push for the access to the food pellet.

Comments on revisions:

We thank the authors for the significant improvement of the English in the revised version and for the replacement of some conceptual terms that now seem more relevant and appropriate. We only noticed that the term "society" remains in use, although it might not be appropriate to describe a mouse colony (see previous review).

Conclusive remarks

Although this protocol aims to provide a novel approach to evaluate social ranks in mice, it is not clear how it really brings a significant advance in neuroscience research. The FPCT dynamic is very similar to the one observed in the tube test, where mice compete to navigate forward in a narrow space, constraining the opponent to go backwards. The main difference between the FPCT and the tube test is the presence of food between the opponents. In the tube test, food reward was initially used to increase motivation to cross the tube and push the opponent upon the testing day. This component has been progressively abandoned, precisely because it was not necessary for the mice to compete in the tube.

This paper would really bring a significant contribution to the field by providing a neuronal imaging or manipulation correlate to the behavioral outcome obtained by the application of the FPCT.

<https://doi.org/10.7554/eLife.103748.2.sa3>

Reviewer #2 (Public review):

Summary:

In this study, the authors have devised a novel assay to measure relative social rank in mice that is aimed at incorporating multiple aspects of social competition while minimizing direct contact between animals. Forming a hierarchy often involves complex social dynamics related to competitive drives for different fundamental resources, including access to food,

water, territory and sexual mates. This makes the study of social dominance and its neural underpinnings hard, warranting the development of new tools and methods that can help understand both social function as well as dysfunction.

Strengths:

This study showcases an assay called the Food Pellet Competition Test, where cagemate mice compete for food, without direct contact, by pushing a block in a tube from opposite directions. This task ran with stranger mice leads to more variable outcomes, suggesting co-housing helps stabilize outcomes. The authors have attempted to quantify motivation to obtain the food independent of other factors by running the assay under two conditions: one where the food is accessible and one where it isn't. This assay results in high outcome consistency across days for females and males paired housed and for male groups of three. Further, the determined social ranks correlate strongly with two common assays: the tube test and the warm spot test.

Weaknesses:

This new assay has limited ethological validity since mice do not compete for food without touching each other with a block in the middle. In addition, the assay may only be valid for a single trial per day, making its utility for recording neural recordings and manipulations limited to a single sample per mouse. The authors claim, as currently stated in results, for the new control experiment in 1H-J is not warranted given that 6/8 mice had majority winning or losing across all strangers.

<https://doi.org/10.7554/eLife.103748.2.sa2>

Reviewer #3 (Public review):

Summary:

The laboratory mouse is an ideal animal to study the neural and psychological underpinnings of social dominance behavior because of its economic cost and the animals' readiness to display dominant and subordinate behaviors in simple and testable environments. Here, a new and novel method for measuring dominance and the individual social status of mice is presented using a food competition assay. Historically, food competition assays have been avoided because they occur in an open arena or the home cage, and it can be difficult to assess who gets priority access to the resource and to avoid aggressive interactions such as bite wounding. Now, the authors have designed a narrow rectangular arena separated in half by a sliding floor-to-ceiling obstacle, where the mice placed at opposite sides of the obstacle compete by pushing the obstacle to gain priority access to a food pellet resting on the arena floor under the obstacle. One can also place the food pellet within the obstacle to restrict priority access to the food and measure the time or effort spent pushing the obstacle back and forth. As hypothesized, the outcomes in the food competition test were significantly consistent with those of the more common tube test (space competition) and warm spot competition test. This suggests that these animals have a stereotypic dominance organization that exists across multiple resource domains (i.e., food, space, and temperature). Only male and female C57 mice in same-sex pairs or triads were tested.

Strengths:

The design of the apparatus and the inclusion of females are significant strengths within the study.

Weaknesses:

There are at least two major weaknesses of the study: the test with unfamiliar non-cagemates and not providing the mice time to recognize who they are competing with.

The authors conclude in the first section of the results that they "did not detect significant difference in winning/losing results between unfamiliar non-cagemate male mice." Given the data and analysis provided, I believe this statement is false. My understanding is that the authors would like to show that the establishment of social relationships (i.e., familiarity) is necessary for FPCT to distinguish social ranks of mice. There are many ways to test this. The simplest would be to randomly pair unfamiliar non-cagemates that are housed in isolation with one another and see if they perform at chance, individually. The more involved empirical way would be to measure the ranks of mice in a social group, then test them with unfamiliar non-cagemate mice to see if they maintain their outcomes regardless of social familiarity, or return to chance outcomes when paired with non-cagemates. Figure 1I clearly shows that they did not perform at chance. Since the outcome is win or lose, then the probability of getting all of one outcome 4 times in a row would be 1 in 16. The data shows that this occurred twice, so 2 mice of 8 had the same outcome 4 times in a row (i.e., Mouse B3 and A1). So, they did not perform at chance. I am not even sure if there are enough animals here to test this question. One may need to consult a mathematician. Moreover, the original tube-test study by Lindzey et al. 1961 (<https://www.nature.com/articles/191474a0>) used unfamiliar non-cagemate male mice, and showed that 100% of the A/alb strain won more than half of their oppositions against C3H and DBA/8 mice. Thus, A/alb mice were more "dominant" mice relative to C3H or DBA/8. Taking into consideration the results, is mouse A1 naturally dominant? So maybe it doesn't matter what mouse you pair with it, it will always win? If this is true, is "individual identification of the partner" actually necessary to get this outcome? All they have to do is push to get the food reward, does it matter who is on the other side? If one wants to measure social dominance relationships, then it should matter who is on the other side. If one would like to measure attributes of dominant behavior (e.g., pushing), then one may do so and not insinuate a social link. Studying dominance relationships (i.e., social ranking) of animals is an extremely difficult task. We must ensure that we are not assigning something about a relationship that does not exist. Please read "Dominance: The baby and the bathwater" but Irwin Bernstein, <https://annas-archive.org/scidb/10.1017/s0140525x00009614/>

Unlike the tube test and warm spot test, the food competition test presented here provides no opportunity for the animals to identify their opponent. That is, they cannot sniff their opponent's fur or anogenital region, which would allow them an opportunity to identify them individually. Thus, as the authors state, the test only measures a psychological motivation to get a food reward. Notably, the outcome in the direct and indirect testing of food competition is in agreement, leaving many to wonder whether they are measuring the social relationship or the effort an individual puts forth in attaining a food reward regardless of the social opponent. Specifically, in the direct test, an individual can retrieve the food reward by pushing the obstacle out of the way first. In the indirect test, the animals cannot retrieve the reward and can only push the obstacle back and forth, which contains the reward inside. In Figure 2F, you can see that winners spent more time pushing the block in the indirect test—albeit not significantly. Thus, whether the test measures a social relationship or just the likelihood to gain priority access to food is unclear. To rectify this issue, the authors could provide an opportunity for the animals to interact before lowering the obstacle and raising(?) a food reward. They may also create a very long one-sided apparatus to measure the amount of effort an individual mouse puts forth in the indirect test with only one individual—or any situation with just one mouse where the moving obstacle is not pushed back, and the animal can just keep pushing until they stop. This would require another experiment. It also may not tell us much more since it remains unclear whether inbred mice can individually identify one another (see <https://doi.org/10.1098/rspb.2000.1057> for more details).

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors present a new protocol to assess social dominance in pairs and triads of C57BL/6j mice, based on a competition to access a hidden food pellet. Using this new protocol, the authors have been able to identify stable ranking among male and female pairs, while reporting more fluctuant hierarchies among triads of males. Ranking readouts identified with this new apparatus were compared to the outcomes obtained with the same animals competing in the tube and in the warm spot tests, which have been both commonly used during the last decade to identify social ranks in rodents under laboratory conditions.

Strengths:

FPCT allows for easy and fast identification of a winner and a loser in the context of food competition. The apparatus and the protocol are relatively easy and quick to implement in the lab and free from any complex post-processing/analysis, which qualifies it for wide distribution, particularly within laboratories that do not have the resources to implement more sophisticated protocols. Hierarchical readouts identified through the FPCT correlate with social ranks identified with the tube and the warm spot tests, which have been widely adopted during the last decade and allow for study comparison.

Weaknesses:

While the FPCT is validated by the tube and the warm spot test, this paper would have gained strength by providing a more ethologically based validation. Tube and warm spot tests have been shown to provide conflicting results and might not been a sufficient measurement for social ranking (see Varholik et al, Scientific reports, 2019; Battivelli et al, Biological psychiatry, 2024). Instead, a general consensus pushing toward more ethological approaches for neuroscience studies is emerging.

We appreciate all the reviewers for recognizing the strength of the FPCT setup and the data. We also appreciate the reviewers for pointing out weakness and giving us valuable suggestions that help us to improve the quality of our manuscript through revision.

In this manuscript, we found the ranking results of the FPCT were largely consistent with the tube and the warm spot tests. Such a finding was unexpected by us as we considered that different competitive targets of different paradigms should provide the mice with distinct appeals and enable them to exert their specific advantages. However, the consistency between the FPCT and tube test was observed in the pairs of female mice, pairs of male mice and triads of male mice. The consistency between the FPCT, tube test and warm spot test was observed in pairs of male mice and triads of male mice. Thus, we concluded that there is a social rank-order stability of mice.

We acknowledge that it'd better if this conclusion could be validated by more ethological approaches like urine-marking analysis and water competition test. Whereas, we did not rule out inconsistency of ranking results between two or more paradigms. Actually, there were inconsistent cases in our experiments. The inconsistency of ranking results between paradigms, even between FPCT and tube test, could be amplified if the tests were operated

with other details of experimental protocols and conditions. This is in that too many factors and aspects can affect the readouts, such as formation of colony, tasks, test protocols, habituation and training. Using tube test itself, both stable 1,2 and unstable 3 ranking results have been reported.

Other papers already successfully identified social ranks dyadic food competition, using relatively simple scoring protocol (see for example Merlot et al., 2006), within a more naturalistic set-up, allowing the 2 opponents to directly interact while competing for the food. A potential issue with the FPCT, is that the opponents being isolated from each other, the normal inhibition expected to appear in subordinates in the presence of a dominant to access food, could be diminished, and usually avoiding subordinates could be more motivated to push for the access to the food pellet.

The hierarchical structure of mice colony could be established on the basis of physical aspects—such as muscular strength, vigorousness of fighting—and psychological aspects—such as boldness, focused motivation, active self-awareness of status. In the contexts of currently available food contest paradigms where the mice compete with bodily interaction, the physical and psychological aspects are intermingled in the interpretation of the mice's winning/losing. In the FPCT, the opponents are isolated from each other so that the importance of direct bodily interaction in a competition is minimized, facilitating the exposure of psychological factors contributing to the establishment and/or expression of social status of the mice. In this study, the overall stable ranking results across the FPCT, tube test and warm spot test indicate that the status sense of animals is part of a comprehensive identify of self-recognition of individuals in an established mice social colony.

There are issues with use of the English language throughout the text. Some sentences are difficult to understand and should be clarified and/or synthesized.

We thank the reviewer for pointing out language issues. We have carefully corrected the grammar errors.

Open question:

Is food restriction mandatory? Palatable food pellet is not sufficient to trigger competition? Food restriction has numerous behavioral and physiological consequences that would be better to prevent to be able to clearly interpret behavioral outcomes in FPCT (see for example Tucci et al., 2006).

We thank the reviewer for raising this question. In the preliminary experiments, we noticed that food restriction was mandatory and palatable food pellet was not sufficient to trigger competition. In order to limit the potential influence of food restriction on competitive behavior, the mice underwent only a 24-hour food deprivation period at the beginning of training, followed by mild restriction of food supply to meet basic energy requirement.

Conclusive remarks:

Although this protocol attempts to provide a novel approach to evaluate social ranks in mice, it is not clear how it really brings a significant advance in neuroscience research. The FPCT dynamic is very similar to the one observed in the tube test, where mice compete to navigate forward in a narrow space, constraining the opponent to go backward. The main difference between the FPCT and the tube test is the presence of food between the opponents. In the tube test, a food reward was initially used to increase motivation to cross the tube and push the opponent upon the testing day. This component has been progressively abandoned, precisely because it was not necessary for the mice to compete in the tube.

This paper would really bring a significant contribution to the field by providing a neuronal imaging or manipulation correlate to the behavioral outcome obtained by the application of the FPCT.

Thank the reviewer for this comment on the significance of the FPCT paradigm. In this manuscript, we think it is interesting to report that the ranking results were consistent across the FPCT, tube test and warm spot test. This finding indicates that the status sense of animals might be a part of a comprehensive identify of self-recognition of individuals in an established social colony.

Moreover, we are conducting researches on biological consequences and mechanisms of social competition. Hopefully, the results of the on-going project will be published in the near future.

Reviewer #2 (Public review):

Summary:

In this study, the authors have devised a novel assay to measure relative social rank in mice that is aimed at incorporating multiple aspects of social competition while minimizing direct contact between animals. Forming a hierarchy often involves complex social dynamics related to competitive drives for different fundamental resources including access to food, water, territory, and sexual mates. This makes the study of social dominance and its neural underpinnings hard, warranting the development of new tools and methods that can help understand both social functions as well as dysfunction.

Strengths:

This study showcases an assay called the Food Pellet Competition Test where cagemate mice compete for food, without direct contact, by pushing a block in a tube from opposite directions. The authors have attempted to quantify motivation to obtain the food independent of other factors such as age, weight, sex, etc. by running the assay under two conditions: one where the food is accessible and one where it isn't. This assay results in an impressive outcome consistency across days for females and males paired housed and for male groups of three. Further, the determined social ranks correlate strongly with two common assays: the tube test and the warm spot test.

Weaknesses:

This new assay has limited ethological validity since mice do not compete for food without touching each other with a block in the middle. In addition, the assay may only be valid for a single trial per day making its utility for recording neural recordings and manipulations limited to a single sample per mouse. Although the authors attempt to measure motivation as a factor driving who wins the social competition, the data is limited. This novel assay requires training across days with some mice reaching criteria before others. From the data reported, it is unclear what effects training can have on the outcome of social competition. Beyond the data shown, the language used throughout the manuscript and the rationale for the design of this novel assay is difficult to understand.

We appreciate the reviewers for the valuable comments on the strength and weakness of our manuscript.

The design mentality of the FPCT was to (1) provide researchers with a choice of new food competition paradigm and (2) expose psychological factors influencing the establishment

and/or expression social status in mice by avoiding direct physical competition between contenders (see revised Abstract and the last paragraph in the Introduction).

As a result, the consistent ranking across the FPCT, tube test and warm spot test might indicate that the status sense of animals is part of a comprehensive identify of self-recognition of individuals in an established social colony.

We suggest to perform the FPCT test one trial per day per mouse as the mice might lose interest in the food pellet if it is tested frequently in a day, but it is practical to perform the FPCT assay for several days.

Regarding the training, we suggest 4-5 days for training as we did. In this revision, we add training data which show the progressing latency of food-getting of mice (Figure 1). At the last day of training, the mice would go directly to push the block and eat the food after they entered the arena.

We thank the reviewer for pointing out language issues. We have carefully corrected the errors.

Reviewer #3 (Public review):

Summary:

The laboratory mouse is an ideal animal to study the neural and psychological underpinnings of social dominance behavior because of its economic cost and the animals' readiness to display dominant and subordinate behaviors in simple and testable environments. Here, a new and novel method for measuring dominance and the individual social status of mice is presented using a food competition assay. Historically, food competition assays have been avoided because they occur in an open arena or the home cage, and it can be difficult to assess who gets priority access to the resource and to avoid aggressive interactions such as bite wounding. Now, the authors have designed a narrow rectangular arena separated in half by a sliding floor-to-ceiling obstacle, where the mice placed at opposite sides of the obstacle compete by pushing the obstacle to gain priority access to a food pellet resting on the arena floor under the obstacle. One can also place the food pellet within the obstacle to restrict priority access to the food and measure the time or effort spent pushing the obstacle back and forth. As hypothesized, the outcomes in the food competition test were significantly consistent with those of the more common tube test (space competition) and warm spot competition test. This suggests that these animals have a stereotypic dominance organization that exists across multiple resource domains (i.e., food, space, and temperature). Only male and female C57 mice in same-sex pairs or triads were tested.

Strengths:

The design of the apparatus and the inclusion of females are significant strengths within the study.

Weaknesses:

There are at least two major weaknesses of the study: neglecting the value of test inconsistency and not providing the mice time to recognize who they are competing with.

Several studies have demonstrated that although inbred mice in laboratory housing share similar genetics and environment, they can form diverse types of hierarchical organizations (e.g., loose, stable, despotic, linear, etc.) and there are multiple resource domains in the home cage that mice compete over (e.g., space, food, water, temperature, etc.). The advantage of using multiple dominance assays is to understand the nuances of

hierarchical organizations better. For example, some groups may have clear dominant and subordinate individuals when competing for food, but the individuals may "change or switch" social status when competing for space. Indeed, social relationships are dynamic, not static. Here, the authors have provided another test to measure another dimension of dominance: food competition. Rather than highlight this advantage, the authors highlight that the test is in agreement with the standard tube test and warm spot test and that C57 mice have stereotypic dominance across multiple domains. While some may find this great, it will leave many to continue using the tube test only (which measures the dimension of space competition) and avoid measuring food competition. If the reader looks at Figures 6E, F, and G they will see examples of inconsistency across the food competition test, tube test, and warm spot test in triads of mice. These groups are quite interesting and demonstrate the diversity of social dynamics in groups of inbred mice in highly standardized environmental conditions. Scientists interested in dominance should study groups that are consistent and inconsistent across multiple dimensions of dominance (e.g., space, food, mates, etc.).

Unlike the tube test and warm spot test, the food competition test presented here provides no opportunity for the animals to identify their opponent. That is, they cannot sniff their opponent's fur or anogenital region, which would allow them an opportunity to identify them individually. Thus, as the authors state, the test only measures psychological motivation to get a food reward. Notably, the outcome in the direct and indirect testing of food competition is in agreement, leaving many to wonder whether they are measuring the social relationship or the effort an individual puts forth in attaining a food reward regardless of the social opponent. Specifically, in the direct test, an individual can retrieve the food reward by pushing the obstacle out of the way first. In the indirect test, the animals cannot retrieve the reward and can only push the obstacle back and forth, which contains the reward inside. In Figure 4E, you can see that winners spent more time pushing the block in the indirect test. Thus, whether the test measures a social relationship or just the likelihood of gaining priority access to food is unclear. To rectify this issue, the authors could provide an opportunity for the animals to interact before lowering the obstacle and raising(?) a food reward. They may also create a very long one-sided apparatus to measure the amount of effort an individual mouse puts forth in the indirect test with only one individual - or any situation with just one mouse where the moving obstacle is not pushed back, and the animal can just keep pushing until they stop. This would require another experiment. It also may not tell us much more since it remains unclear whether inbred mice can individually identify one another

(see <https://doi.org/10.1098/rspb.2000.1057> for more details).

A minor issue is that the write-up of the history of food competition assays and female dominance research is inaccurate. Food competition assays have a long history since at least the 1950s and many people study female dominance now.

Food competition: <https://doi.org/10.1080/00223980.1950.9712776>, [https://psycnet.apa.org/fulltext/1953-03267-](https://psycnet.apa.org/fulltext/1953-03267-001.pdf)

[001.pdf, https://doi.org/10.1016/j.bbi.2003.11.007](https://doi.org/10.1016/j.bbi.2003.11.007), <https://doi.org/10.1038/s41586-02204507-5>

Female dominance: history <https://doi.org/10.1016/j.cub.2023.03.020>, [https://doi.org/10.1016/S031-9384\(01\)00494-2](https://doi.org/10.1016/S031-9384(01)00494-2), <https://doi.org/10.1037/0735-7036.99.4.411>

We thank the reviewers very much for so many helpful comments and suggestions.

In this manuscript, we want to address the overall and averaged consistency of ranking results between FPCT, tube test and warm spot test) as an unexpected finding. We agree that

the inconsistency of social ranking occurred between trials and between paradigms should not be ignored. In the revision, we added description and discussion of inconsistent part of the different test paradigms (paragraph 2 in the section 3 of the Result, last 2 sentences of paragraph 4 in the Discussion)

Although the two opponents were separated each other, they were able to see and sniff each other because the block is transparency, there are holes in the lower portion of the block, and there is the gap between the block and chamber (Supplementary figures 1 and 2). In the female but not male groups, the presence of a cagemate opponent during the test 1 could significantly disturb the female mice and increase the its latency to get the food, comparing with last day of training when there was no opponent (Figure 3A). This indicates that one mouse, at least female mouse, could identify the existence of the opponent in the opposite side of the chamber. To further see whether social relation was influential to readouts of the FPCT, we performed additional experiments using two groups of non-cagemate mice to perform the competition. We did not detect obviously different ranks between the two groups (Figure 1H-1J), suggesting that establishment of social colony is necessary for FPCT to distinguish social ranks of mice.

Thank the reviewer for reminding us to recognize the history of food competition assays. We have added the citations and discussions of related literatures, both for male (paragraph 2 in the Introduction; paragraph 3 in the Discussion) and female (paragraph 1 of section 3 in the Results; paragraph 4 in the Discussion) mice.

Reviewer #1 (Recommendations for the authors):

There are issues with use of the English language throughout the text. Some sentences are difficult to understand and should be clarified and/or synthesized.

We appreciate the reviewer for constructive comments and helpful corrections.

"Despite that 6 in 9 groups of mice display some extent of flipped ranking (Figures 6B-6G) and only 3 in 9 groups displayed continuously unaltered ranking (Figure 6H) during a total of 9 trials consisting of 3 trials of FPCT, 3 trials of tube test and 1 trial of WST, an obvious stable linear intragroup hierarchy was observed throughout all the trials and tasks"

The above sentence has been re-written as: The ranking result showed that 6 in 9 groups of mice displayed some extent of flipped ranking (Figures 4B-4G), and only 3 in 9 groups displayed continuously unaltered ranking (Figure 4H). Averagely, in the totally 27 trials consisting of 12 trials of FPCT, 12 trials of tube test and 3 trials of WST, an obvious stable linear intragroup hierarchy was observed across all the trials and tasks (paragraph 1 of section 4 in the Results).

"it is hard to attribute winning a competition in a shared space to stronger motivation rather than muscular superiority".

The above sentence has been deleted and re-written in paragraph 1 of section 4 in the Results and paragraph 3 in the Discussion.

"Unexpectedly, in most of the trials the mice preserved the winner or loser identity acquired in FPCT into tube test and WST (Figures 5L-5O)".

Why this is unexpected? Instead, it looks like this result is expected (tube test has been successfully applied to identify ranks in females, see Leclair et al, eLife, 2021).

We thank the reviewer for raising this point. FPCT is different from tube test and warm spot test at least in two aspects: competition for food vs space; presence vs absence of direct bodily interaction during competition. Some mice might be active in food competition, but not in space competition, while others might be on the contrary. Some mice might be good at physical contest, while others might be good at play tricks. Therefore, these factors made us expect task-specific outcomes of ranking results.

Vocabulary issues:

"Stereotypic", to talk about rank stability in a different context does not look appropriate. In behavioral neuroscience, stereotypy is more expected to intend abnormal repetitive behaviors. The stability that the authors seem to indicate with the word "stereotype" refers rather to the concept of "consistency" or "stability".

We thank the reviewer for this detailed explanation. We have chosen to use "stability" to describe the data.

"Society", to talk about groups or colonies of animals sounds a bit odd. Society evokes more abstract concepts more likely to fit with human organization. I suggest the use of "group" or "colony".

"Hide" to qualify the block preventing access to the food pellet. It is said that the block is transparent. We suggest the use of "inaccessible" instead of hidden.

We strongly encourage the authors to further edit the entire script to improve language.

Thank the reviewer for kind correction. We have corrected the above vocabulary misuse.

Technical issues / typos:

Figure 1. The picture does not seem optimal to visualize the apparatus.

Missing unit legend in Figure 4E.

Supplementary videos 2 and 4 are missing.

We have added a frontal view of the apparatus in the figure (Supplementary Figure 1), added a unit to the Figure 2F (previous Figure 4E), and we will make sure to upload the missing videos.

Reviewer #2 (Recommendations for the authors):

While the assay shows promise as a tool for studying social dominance, the study suffers from some limitations such as lack of ethological relevance. In addition, there is a lack of rationale and methodological clarity in the manuscript that can impact the ability of other scientists to be able to perform this novel assay.

(1) Related to lack of scientific rigor:

a. In the first paragraph of the introduction, the authors mention that "disability in social recognition and unsatisfied social status are associated with brain diseases such as autism, depression and schizophrenia". Both papers that they cited refer to mouse models, not humans (which is the species that is attributed these diagnoses clinically). In addition, neither citation discusses schizophrenia. While social dysfunctions can indeed be related to these diseases, to my knowledge this is not caused by a change in "social

status" and there is no human data with patient populations and social status. Therefore, this sentence is inaccurate and there is no research that demonstrates that.

We thank the reviewer for raising this point. To express the opinion and cite literatures more accurately, we improved the sentence in the 1st paragraph of Introduction as follows: "Impaired awareness of social competition has been documented in individuals with autism spectrum disorder (ASD)^{4,5}, and reduced social interaction has been characterized in corresponding animal models⁶. Similarly, maladaptive responses to social status loss has been associated with patient depressive disorders^{7,8} and animal models of depression^{1,9}". The reviewer is right that no patient disease is causally related with social status, and only depression has been proposedly associated with change of social status^{7,8}.

b. In the second paragraph of the introduction, the authors mention a scarcity of research papers with designs for food competition-based social hierarchy assays for mice. At least two such papers have been published in the past few years (DOIs <https://doi.org/10.1038/s41586-021-04000-5> and <https://doi.org/10.1038/s41586-022-04507-5>). The authors should acknowledge the existence of these and other assays and discuss how their work would be related. In the same paragraph, they also mention that existing assays suffer from "hierarchy instability" and "complex calculations" without showing any citations or details for these claims.

021-04000-5 and <https://doi.org/10.1038/s41586-022-04507-5>). The authors should acknowledge the existence of these and other assays and discuss how their work would be related. In the same paragraph, they also mention that existing assays suffer from "hierarchy instability" and "complex calculations" without showing any citations or details for these claims.

We thank the reviewer for raising this point. We acknowledged that there are some available food competitions to measure social hierarchy for mice. But relative to space competition, food competition tests have not been used so commonly and widely. No food competition paradigm has been accepted as generally as some space competition paradigms like tube test and warm spot test. To improve the language and scientific expression, we revised the sentences as follows: "Relative to space competition, food competition tests for mice have been designated and applied less commonly in animal studies despite its long history²⁸⁻³⁰. Several issues could be thought to be the underlying limitations for the application of food competition paradigms. First, there are methodological issues in some of these approaches, such as long video recording duration and difficulty in analyzing animal's behaviors during competitive physical interaction in videos, hindering their application by laboratories that cannot afford sophisticated equipment and analysis". Corresponding citations have been updated (see paragraph 3 in the Introduction).

c. The authors say that their study is the first to demonstrate that female mice follow social ranks. This is not the first study to do so and the authors should acknowledge existing publications that have done the same (eg DOI <https://doi.org/10.7554/eLife.71401>).

We have followed the reviewer's suggestion to increase citations regarding social ranking of female mice tested by competition paradigms, especially food competition paradigms (see paragraph 1 of section 3 in the Results; paragraph 4 in the Discussion).

(2) Related to problems with interpretation of data:

a. The authors showed the assay works for females and males in pairwise housing, but two mice don't make a hierarchy, as hierarchies require a minimum of three individuals. Therefore, whether the assay works for females caged in three is an important question that is unaddressed in this study and is a caveat. extended the competition assay to male mice that are housed in cages of three. It would be important to show whether the assay generalizes well for female mice with this three-animal housing as well as discuss the effect of using even bigger groups of mice on the results of the assay.

We thank the reviewer for raising questions related to the interpretation of data and giving us the insightful the suggestions. We agree that it is interesting and important to probe if FPCT works for a group of three female mice. Although social rankings of pairs of male and female mice were not significantly different (new Figure 2D-2F and 3F-3H), that of triads of male and female mice could be different. We have tested trades of male mice and found that the mice displayed an overall linear hierarchical ranking. We would like to use FPCT to investigate the rankings of trades of female mice and even bigger group of mice in the future. In the present manuscript we'd like to address the feasible application of the FPCT in smaller groups. In the Discussion, we add contents commenting group size effect on social competition tests (see paragraph 4 in the Discussion).

b. The authors claim that "test 2" of their assay helps assert the motivation of mice for social competition as in Figure 4E. This could simply be a readout of how strong the mice are (muscle mass). To claim that this is indeed related to motivation during the FPCT assay, the authors should show the correlation of this readout with the latency to push the block during the social competition task.

We appreciate the reviewer for raising this question. The dimensions establishing the social structures include physical and psychological factors. In the FPCT paradigm, the two contenders are separated so that physical factors are minimized in this context and psychological factors should play more important role in competition in comparison with previous reported food competition paradigms. Therefore, in the revised manuscript we consider to attribute the ranking results mainly to psychological factors, rather than only motivation which is just one of the numerous psychological factors (paragraph 3 of Discussion). Moreover, in the Discussion we point out that we could not exclude physical factors still participate in the determination of competitive outcomes since some of mice pairs pushed the block simultaneously (paragraph 3 of Discussion).

c. The authors mention that they are interested to understand which factors lead to the outcome of the competition such as age, sex, physical strength, training level, and intensity of psychological motivation. However, in all their runs of the assay, they always matched these variables between the competitors. They should clarify that they were instead controlling for these variables. Another thing to note here is that while they controlled the body mass of the animals, that isn't the same as physical strength, as a lighter mouse can have more muscle mass than a heavier mouse. They should either specify this limitation or quantify the additional metric of "muscle mass" which is a much better proxy for physical strength. Thus, the claim that the outcome of the competition is solely affected by motivation is not convincing since they didn't rule out the others such as quantifying the rate of learning during training and strength.

We thank the reviewer for addressing this question. As our response to the question in (c), we acknowledge that it is not accurate to ascribe the outcomes of FPCT to psychological motivation. In the revised manuscript, the dimensions of contributing factors to the outcomes of FPCT have been simplified to physical and psychological factors. We consider that the psychological factor could be the main driver of mice participating in FPCT (see paragraph 3 of Discussion).

d. In the discussion, the authors mention that their task only requires a single day of food deprivation (the day before the first trial) while other assays suffer from a continued food deprivation protocol. However, the authors also use 10g per cage as the amount of food instead of giving them ad libitum access. Limited food is a food deprivation method. Thus, this is an inaccurate claim.

We thank the reviewer for raising this point. We have clarified the requirement of food restriction for FPCT in the revision. The mice were deprived of food for 24 hours while water consumption remained normally to enhance the appeal of the food pellet to the mice. Then, after 24 hours of food deprivation, each cage of mice was given 10 g of food every morning to meet their daily food requirements until the end of the test (see FPCT procedure section in Methods and materials).

e. In the second section of the results, the authors run their assay with female mice that are housed in cages of two. This section suffers from the same limitations as the first and can be improved by showing the training data, correlations of competition outcome with "motivation" and ruling out the other factors that could contribute to the outcome. Further, the authors saying that their FPCT assay is enough to show that female mice follow a social hierarchy by itself is a weak claim. They should instead include their cross-validation with the others to strengthen it.

We appreciate the reviewer for raising this question. We have taken the reviewer's suggestion to show the training data (Figures 1E, 2A and 3A). As the factors contributing to the outcomes of FPCT are diverse, we'd like not to control and determine the exact factor in the current manuscript. We agree with the reviewer that cross-validation with different paradigms is suggested for the studies to rank social hierarchy as the ranking results could be variable with tasks, procedures and operations.

f. In the last paragraph of the introduction, the authors mention how their assay involves "peaceful competition" since the mice are not in direct contact and hence cannot exhibit aggression. The authors do not address the limitation that a lack of physical contact actually makes the assay less ethological. Further, since the mice are housed in groups of two and three, it is not guaranteed that the mice will not be aggressive during their time in the home cage, which could affect their behavior during the competition assay. Whether the assay causes more aggression in the cage due to the lack of physical contact during the competition is not addressed in this study.

We thank the reviewer for raising this point. Diverse factors affect the outcomes of a food competition test, some of which belong to psychological factors and others belong to physical factors. We agree that a lack of physical contact makes the assay less naturally ethological. However, when the social statuses have been established during habituation housing a group of mice for enough time, the win/lose outcomes in the FPCT could be a readout of the expression of social statuses since the mice cannot exhibit aggression in the test. We have revised the Introduction and Discussion (paragraph 3 of Discussion). Thank you.

(3) Related to lack of methodological rigor and rationale clarity:

a. In the first section of the results, the authors run their assay with male mice that are housed in cages of two. While the data that they display is promising, we do not see how mice change behavior across days of training and how that relates to the outcome of the competition. It would be valuable to also show the training data for the mice, answering questions related to competency and any inter-animal variabilities prior to rank assessment. Plotting the training data across all days would be helpful for the other parts of the results as well. This is especially important because the methods mention that mice are trained until they get to the criterium, so this means that different individuals get different amounts of training.

We appreciate the reviewer for addressing the importance of showing training data. We have taken the reviewer's suggestion and shown the training data (Figures 1E, 2A and 3A).

b. It is unclear why the assay was run only once per mouse pair per day since most protocols for the tube test involve multiple repetitions each day while alternating the side from which the mice enter. The authors should address whether a single trial per day is enough to show consistent results and that it wouldn't vary with more.

We suggest to run the FPCT once or twice per mouse per day under conditions of mild food restriction, training and test procedures in this manuscript. Frequent tests might make the mice's interest in the food pellet gradually diminished because the food supply was not fully deprived. According to our data, the outcomes of FPCT in 4 consecutive days were overall stable.

c. In the results the authors say that they "raised 3 male mice" which may be incorrect because they report in the methods buying the mice buy mice and they housed all their mice for only three days before running the assay which might be too little for the hierarchy to stabilize. The authors should comment on what was the range of the cohabitation across different cages and whether it had an impact on the results.

According to our experiments, housing the mice for 3 days is enough to establish a mice social colony with relative stable status structure. Prolonged housing may produce either similar, stabler or more dynamic social colony.

d. There are also some formatting and/or convention issues in the results. The first figure callout in the results is for Figure 4 instead of Figure 1 (which is the standard). This is because the authors do not explain how the mice are trained for the task in the results section and show limited data about the training of the task. Not showing comprehensive training data would make replication of this study very difficult.

We appreciate the reviewer for raising this question. We have re-arranged the figures. The new arrangement of figures started with schematic drawing of FPCT procedure and training data (Figure 1).

e. The authors don't report the exact p-values in the figures

We reported the difference level in the figures in the revised manuscript. Thank you.

4. The writing of the manuscript suffers from a lack of clarity in most sections of the manuscript.

Here are several examples that are critical:

a. In the title and abstract, it isn't clear what the authors mean by "stereotype". It could be a behavior during the competition, or that the social ranks across assays are correlated or that the rank for the new assay is consistent across days.

b. There are several instances where the authors anthropomorphize mice using human features such as "urbanization" and "society" which are not established factors affecting mouse hierarchy. This further extends to anthropomorphizing mice in ways that are not standard such as an animal being "timid" or "bold" which would be hard to measure in mice, if not impossible.

c. Across the social dominance literature, relative social rank is described using more general "dominant" and "subordinate" titles instead of "superior" and "inferior" that are sometimes used in the manuscript. The authors should follow the standard language so that readers understand.

d. In the third paragraph of the introduction, the authors say "Thus, it is more likely expected that different paradigms to weigh the social competency and status may lead to diverse readouts, given that competitive factors are included in competition paradigms." This sentence suffers from multiple syntax errors thereby reducing clarity

e. There are several typos in the manuscript such as using "dominate" instead of "dominant", "grades" instead of "outcomes" and "forth" instead of "fourth", to give a few examples.

We thank the reviewer for careful reading of the manuscript and very helpful comments. We have taken the above suggestions and improved the writing of the manuscript. For examples, "stereotype" was replaced by "stability", mice "society" was expressed by "colony", the sentence "Thus, it is more.... in competition paradigms" has been deleted.

Reviewer #3 (Recommendations for the authors):

(1) The justification for the design of this new test paradigm is unclear. In the abstract, you state that the field needs a reliable, valid, and easily executable test. Your test provides this, as you state, but how is it better than the tube test? Does the tube test suffer from task-specific win-or-lose outcomes? Can you provide evidence for this? The nature methods protocol for the tube test (<https://doi.org/10.1038/s41596-018-0116-4>) "strongly suggest using more than two dominance measures, for example, by also carrying out the warm spot test, or territory urine marking or ultrasonic courtship vocalization assays." This would suggest that results from the tube test can be task-specific, but I am not convinced that you have demonstrated that results from your food competition test are not task-specific. Indeed, by your title, one must run multiple tests.

This same problem is apparent in the introduction. In the second paragraph, there is a discussion of the tube test, warm spot test, and food competition tests. What is the problem with these tests?

I believe that social dominance relationships are complex and dynamic social relationships indicating who has priority access to a resource between multiple animals that live together. In these living situations, several resources can often be capitalized competed over-for example, space, food, mates, temperature, etc. Currently, we have tests to measure space via the tube test or urine marking, mates via ultrasonic vocalization, temperature via warm spot test, and food via food competition assays. The tube test, urine marking assay, and ultrasonic vocalization test have been demonstrated to be reliable, valid, and easily executable. However, the food competition assays are often difficult to execute because it is difficult to interpret the dominant behaviors and aggressive behaviors like bite wounding can occur during the test. Here, you present a new food competition assay to address these issues and show that it can be used in conjunction with other assays to measure social dominance across multiple resources easily. In doing so, you revealed that many same-sex groups of C57 mice have a stereotypic pattern of dominance behavior when competing across multiple types of resources: space, temperature, and food.

I ask that you please rebut if you disagree with me, and adjust your abstract, introduction, and discussion accordingly.

We thank the reviewer for all the constructive comments. We have adjusted the Abstract, Introduction and Discussion of the manuscript.

We recognize and appreciate the valuable tube test, warm spot test and many other competition tests, including food competitions. Tube test and warm spot test are space

competition tasks. Relative to space competition, food competition tests for mice have been designated and applied less commonly in animal studies. Several issues (such as methodological issue, aggressive behaviors occurring in competition, and prolonged food deprivation) could be thought to be the underlying limitations of the application of food competition paradigms (paragraph 3 in the Introduction). Therefore, we clarify that the justification for the design of FPCT was “to have a new choice of food competition paradigm for mice, and to facilitate the exposure of psychological aspects contributing to the winning/losing outcomes in competitions” (last paragraph in the Introduction).

FPCT is different from tube test and warm spot test at least in two ways. FPCT is food completion task where the mice need no physical contact during competition, while tube test and WST are space competition tasks where the mice need direct physical contact during competition. Therefore, we expected inconsistent evaluation results of competitiveness and rankings if we compared FPCT with typically available competition paradigms—tube test and WST (last paragraph in the Introduction).

(2) The design of the test needs to be described before the results. You can either move the methods section before the results or add a paragraph in the introduction to better describe the test. Here, you can also reference Figures 1 through 3 so that the figures are presented in the order of which they are mentioned in the paper. (It is very confusing that the first reference to a figure is Figure 4, when it should be Figure 1).

We appreciate the reviewer for raising this point and giving us suggestions. We have added a new section (section 1) in the Results. In the revised manuscript, the figures in the Results start with Figure 1 which shows schematic drawing of FPCT procedure, training data and some test results (Figure 1).

(3) The sentence describing Figure 4H. You argue that this shows that the mice are well and equally trained. It also shows that they have the same motivation or preference for the food.

We appreciate the reviewer for this helpful comment. Data in previous Figures 4H and 5I have been presented as new Figures 2A and 3A, respectively, of revised manuscript. These retrospect analysis of training data displayed similar training level of food-getting and craving state for food (Sections 2 and 3 in the Results).

(4) "Social ranking of multiple cagemate mice using FPCT, tube test and WST"

Here, you claim that "comparison of inter-task consistency revealed that the ranks evaluated by FPCT, tube test and WST did not differ from each other...Figure 6K." Okay, however, it is important to discuss the three cases when there wasn't consistency between the tests! Figure 6E-G.

We appreciate the reviewer for raising this point. In the revised manuscript, we add description and discussion of inconsistent part of the different test paradigms (paragraph 2 in the section 3 of the Result, last 2 sentences of paragraph 4 in the Discussion)

(5) Replace all instances of "gender" with "sex". Animals do not have a gender.

(6) Adjust the strain of the mice to C57BL/6JNifdc.

We have replaced "gender" with "sex" and “C57BL/6J” with “C57BL/6JNifdc”. Thank you for your careful correction.

(7) *What is the justification for running the warm spot test for one day and the other tests for four days?*

From the consecutive FPCT and tube test, we already knew that the ranking results were overall stable. This stability was still observed in the day of warm spot test. A bad point for frequent warm spot test is that mice get much stress due to exposure in ice-cold environment. Therefore, we terminated the competition test after only one trial of warm spot test.

(8) *Grammar*

The second sentence of the abstract: ...recognized as a valuable...

Results, sentence after "...was observed (Figure 4G)." it should be "Fourth"

We have corrected these and other grammar errors. We appreciate the reviewers for very careful review and all helpful comments.

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